

Assignment # 1—Solution

CS1005-Discrete Structures

1. a) Solution: Proposition. Hence True
 c) Solution: Proposition. Hence True
 e) Solution: Not a Proposition.
 g) Solution: Proposition. Hence False
 i) Solution: Not a Proposition.
 b) Solution: Proposition. Hence False
 d) Solution: Proposition. Hence False
 f) Solution: Not a Proposition.
 h) Solution: Proposition. Hence True
 j) Solution: Not a Proposition.

2. a) Solution: $P \leftrightarrow Q$
 d) Solution: $P \wedge Q$
 b) Solution: $P \vee (\neg P \wedge Q)$
 e) Solution: $P \oplus Q$
 c) Solution: $P \vee Q$

3. a) Solution: $\neg p \wedge q$: $2 > 5$ and 8 is an even integer. Truth Value: False
 b) Solution: $(p \wedge q) \rightarrow r$: If $2 \leq 5$ and 8 is an even integer then 11 is a prime number. Truth Value: True
 c) Solution: $p \rightarrow (q \vee (\neg r))$: If $2 \leq 5$ then 8 is an even integer or 11 is not a prime number. Truth Value: True
 d) Solution: $(\neg p) \rightarrow (\neg q)$: If $2 > 5$ then 8 is an odd integer. Truth Value: True
 e) Solution: $\neg q \leftrightarrow r$: 8 is not an even integer iff 11 is a prime number. Truth Value: False
 f) Solution: $r \leftrightarrow (q \vee p)$: 11 is a prime number iff 8 is an even integer or $2 \leq 5$. Truth Value: True
 g) Solution: $r \rightarrow \neg p$: if 11 is a prime number then $2 > 5$. Truth Value: False
 h) Solution: $(q \wedge \neg r) \rightarrow p$: If 8 is an even integer and 11 is not a prime number then $2 \leq 5$. Truth Value: True

4. a) Solution: True Statement: p
 p: Smartphone B has the most RAM of these three smartphones.
 b) Solution: True Statement: $p \vee q$ (p OR q)
 First identify propositions:
 p: C has more ROM than B. FALSE q: C has a higher resolution camera than B TRUE
 c) Solution: False Statement: $p \wedge q \wedge r$ (p AND q AND r)
 First identify propositions:
 p: B has more RAM than A. TRUE q: B has more ROM than A. TRUE
 r: B has a higher resolution camera than A. FALSE
 d) Solution: False Statement: $(p \wedge q) \rightarrow r$ (IF (p AND q) THEN r)
 p: B has more RAM than C. TRUE q: B has more ROM than C. TRUE
 r: B has a higher resolution camera than C. FALSE
 e) Solution: False Statement: $(p \leftrightarrow q)$ (p IF AND ONLY IF q)
 p: A has more RAM than B. FALSE q: B has more RAM than A. TRUE

5. a) Solution: False (the largest annual revenue had Acme Computer, 138 billion dollars)
 p: Quixote Media had the largest annual revenue.
 b) Solution: True Statement: $(p \wedge q)$
 p: Nadir Software had the lowest net profit. True q: Acme Computer had the largest annual revenue. True
 c) Solution: True Statement: $(p \vee q)$
 p: Acme Computer had the largest net profit. False q: Quixote Media had the largest net profit. True
 d) Solution: True Statement: $(p \rightarrow q)$
 p: Quixote Media had the smallest net profit. False q: Acme Computer had the largest annual revenue. True
 e) Solution: True Statement: $(p \leftrightarrow q)$
 p: Nadir Software had the smallest net profit. True q: Acme Computer had the largest annual revenue. True

6. i)
 a) Solution: If you have flu then you will miss the final examination.
 b) Solution: You won't miss the final examination if and only if you pass the course.
 c) Solution: If you miss the examination then you will be failing (not passing) the course.
 d) Solution: You have the flu or you miss the final examination or you pass the course.
 e) Solution: If you have the flu then you'll not pass the course or if you miss the final examination then you'll fail (not pass) the course.
 f) Solution: You have the flu and you miss the examination or you will not miss the final examination and you pass the course.
- ii)
 a) Solution: $r \wedge \neg q$
 b) Solution: $p \wedge q \wedge r$
 c) Solution: $r \rightarrow p$
 d) Solution: $p \wedge \neg q \wedge r$
 e) Solution: $(p \wedge q) \rightarrow r$
 f) Solution: $r \leftrightarrow (q \vee p)$

7. a) "p only if q"
 Solution: If you send me an e-mail message, then I will remember to send you the address.
 b) "a sufficient condition for q is p"
 Solution: If you were born in the United States, then you can be a citizen of the United States.
 c) "if p, q"
 Solution: If you keep your textbook, then it will be a useful reference in your future courses.
 d) "q if p"
 Solution: If the Red Wings's goalie plays well, they will win the Stanley Cup.
 e) "p implies q"
 Solution: If you get the job, then you had the best credentials.
 f) "q whenever p"
 Solution: If there is a storm, then beach erodes.
 g) "q is necessary for p"
 Solution: If you can log on to the server, then you must have a valid password.
 h) "q unless $\neg p$ "
 Solution: If you begin your climb too late, then you will not reach the summit.

8. a) Solution: Following ways to express this conditional statement:
- | | |
|-------------------------------------|---|
| "p implies q" | "It is sunny tomorrow implies I will go for a walk in the woods." |
| "if p, q" | "If it is sunny tomorrow, I will go for a walk in the woods." |
| "p only if q" | "It is sunny tomorrow only if I will go for a walk in the woods." |
| "p is sufficient for q" | "It is sunny tomorrow is a sufficient condition for a walk in the woods." |
| "a sufficient condition for q is p" | "A sufficient condition for a walk in the woods is sunny tomorrow." |
| "q if p" | "I will go for a walk in the woods If it is sunny tomorrow." |
| "q whenever p" | "I will go for a walk in the woods whenever it is sunny tomorrow." |
| "q when p" | "I will go for a walk in the woods when it is sunny tomorrow." |
| "q is necessary for p" | "For a walk in the woods is necessary for sunny tomorrow." |
| "a necessary condition for p is q" | "A necessary condition to be sunny tomorrow for a walk in the woods." |
| "q follows from p" | "I will go for a walk in the woods follows it is sunny tomorrow." |
| "q unless $\neg p$ " | "I will go for a walk in the woods unless it is not sunny tomorrow." |

- b) Solution:
 Converse of $p \rightarrow q$: $q \rightarrow p$ "If I go for a walk in the woods then it will be sunny tomorrow."
 Contrapositive of $p \rightarrow q$: $\neg q \rightarrow \neg p$ "If I do not go for a walk in the woods then it will not be sunny tomorrow."
 Inverse of $p \rightarrow q$: $\neg p \rightarrow \neg q$ "If it is not sunny tomorrow, then I will not go for a walk in the woods."

- c) Solution:
 Inverse: If it is not sunny tomorrow, then I will not go for a walk in the woods.
 Inverse of inverse: If it is sunny tomorrow, then I will go for a walk in the woods.

Converse: If I go for a walk in the woods then it will be sunny tomorrow.

Inverse of converse: If I do not go for a walk in the woods then it will not be sunny tomorrow.

Contrapositive: If I do not go for a walk in the woods then it will not be sunny tomorrow.

Inverse of its contrapositive: If I go for a walk in the woods then it will be sunny tomorrow.

9. i)

a) Solution: Jan is not rich or happy.

b) Solution: Carlos will not bicycle and not run tomorrow.

c) Solution: The fan is not slow and it is not very hot.

d) Solution: Akram is not unfit or Saleem is not injured.

e) Solution: $0 > x > 4$

f) Solution: $-2 \geq x > 5$

g) Solution: The fan is fast, and it is very cold.

h) Solution: Akram is fit or Ahmed is healthy.

ii)

a) Solution: Because the one alternative (staying home) precludes the other (going out), "or" is used in the exclusive sense.

b) Solution: You might both fail to make a payment on time and your late payment might be for an incorrect amount. Hence the inclusive "or" is used here.

c) Solution: If both events happen, you won't go on the trip. Hence the inclusive "or" is used here.

d) Solution: It is implied that you won't be served if you fail to wear a shirt and also fail to wear shoes. Therefore, the inclusive "or" is used here.

10. (i)

a) Solution: $a \vee b$

b) Solution: $\neg c \rightarrow \neg b$

c) Solution: $c \rightarrow d$

d) Solution: $\neg a$

(ii) Solution:

$= (a \vee b) \wedge (\neg c \rightarrow \neg b) \wedge (c \rightarrow d) \wedge (\neg a)$

Law of contrapositive

$= (a \vee b) \wedge (b \rightarrow c) \wedge (c \rightarrow d) \wedge (\neg a)$

Law of Hypothetical Syllogism

$= (a \vee b) \wedge (b \rightarrow d) \wedge (\neg a)$

Law of Disjunctive Syllogism

$= (b) \wedge (b \rightarrow d)$

Modus Ponens

$= d$

(iii) Solution:

Inputs				Statement	Contrapositive	Inverse	Converse of inverse
b	c	$\neg b$	$\neg c$	$\neg c \rightarrow \neg b$	$b \rightarrow c$	$c \rightarrow b$	$b \rightarrow c$
T	T	F	F	T	T	T	T
T	F	F	T	F	F	T	F
F	T	T	F	T	T	F	T
F	F	T	T	T	T	T	T

In words Statement:

Contrapositive: If Ali is not enrolled in BS Computer Science, he doesn't study Discrete Structures.

Inverse: If Ali studies Discrete Structures, he is enrolled in BS Computer Science.

Converse of its inverse: If Ali is not enrolled in BS Computer Science, he doesn't study Discrete Structures.

Hence both statements are equivalent.

11. i)

a) Solution:

$\equiv (p \wedge (\neg (\neg p) \wedge q)) \vee (p \wedge q)$

De-Morgan Law

$\equiv (p \wedge (p \wedge \neg q)) \vee (p \wedge q)$

Double Negation

$\equiv ((p \wedge q) \vee (p \wedge \neg q)) \wedge ((p \wedge q) \vee (p))$

Distributive Law

$\equiv ((p \wedge q) \vee (p \wedge \neg q)) \wedge (p)$

Absorption Law

$$\begin{aligned} &\equiv (p \wedge (q \vee \neg q)) \wedge (p) && \text{Distributive Law} \\ &\equiv ((p \wedge T) \wedge (p)) && \text{Negation Law} \\ &\equiv p \wedge p && \text{Idempotent Law} \\ &\equiv p \end{aligned}$$

b) Solution:

$$\begin{aligned} &\neg p \leftrightarrow q \\ &\Leftrightarrow (\neg p \rightarrow q) \wedge (q \rightarrow \neg p) && \text{Biconditional Equivalence} \\ &\Leftrightarrow (\neg \neg p \vee q) \wedge (\neg q \vee \neg p) && \text{Implication Equivalence (x2)} \\ &\Leftrightarrow (p \vee q) \wedge (\neg q \vee \neg p) && \text{Double Negation} \\ &\Leftrightarrow (q \vee p) \wedge (\neg p \vee \neg q) && \text{Commutative} \\ &\Leftrightarrow (\neg \neg q \vee p) \wedge (\neg p \vee \neg q) && \text{Double Negation} \\ &\Leftrightarrow (\neg q \rightarrow p) \wedge (p \rightarrow \neg q) && \text{Implication Equivalence (x2)} \\ &\Leftrightarrow p \leftrightarrow \neg q && \text{Biconditional Equivalence} \end{aligned}$$

c) Solution:

$$\begin{aligned} &(p \wedge q) \rightarrow (p \rightarrow q) \\ &\equiv \neg(p \wedge q) \vee (p \rightarrow q) && \text{Law of Implication} \\ &\equiv \neg(p \wedge q) \vee (\neg p \vee q) && \text{Law of Implication} \\ &\equiv (\neg p \vee \neg q) \vee (\neg p \vee q) && \text{De Morgan's Law} \\ &\equiv (\neg p) \vee (\neg q \vee (\neg p \vee q)) && \text{Associative Law} \\ &\equiv (\neg p) \vee ((\neg p \vee q) \vee \neg q) && \text{Commutative Law} \\ &\equiv (\neg p) \vee (\neg p \vee (q \vee \neg q)) && \text{Associative Law} \\ &\equiv (\neg p) \vee (\neg p \vee \mathbf{T}) && \text{Negation Law} \\ &\equiv (\neg p) \vee (\mathbf{T}) && \text{Domination Law} \\ &\equiv \mathbf{T} && \text{Domination Law} \end{aligned}$$

d) Solution:

$$\begin{aligned} &\neg(p \vee \neg(p \wedge q)) \\ &\Leftrightarrow \neg p \wedge \neg(\neg(p \wedge q)) && \text{De Morgan's Law} \\ &\Leftrightarrow \neg p \wedge (p \wedge q) && \text{Double Negation Law} \\ &\Leftrightarrow (\neg p \wedge p) \wedge q && \text{Associative Law} \\ &\Leftrightarrow F \wedge q && \text{Contradiction} \\ &\Leftrightarrow F && \text{Domination Law and Commutative Law} \end{aligned}$$

ii)

a) Solution: Equivalent b) Solution: Equivalent c) Solution: Not equivalent d) Solution: Not equivalent

12. i)

a) Solution: $\forall x F(x)$ b) Solution: $\forall x [F(x) \rightarrow S(x)]$ c) Solution: $\forall x [(F(x) \wedge S(x) \rightarrow \neg T(x)) \Leftrightarrow \neg \exists x [F(x) \wedge S(x) \wedge T(x)]]$

ii)

a) Solution: $\forall x F(x, \text{Bob})$ b) Solution: $\forall y F(\text{Alice}, y)$ c) Solution: $\forall x \exists y F(x, y)$

d) Solution: $\neg \exists x \forall y F(x, y)$ e) Solution: $\forall y \exists x F(x, y)$

iii)

a) Solution: $\exists x (P(x) \wedge Q(x))$ b) Solution: $\exists x (P(x) \wedge \neg Q(x))$

c) Solution: $\forall x (P(x) \vee Q(x))$ d) Solution: $\neg \exists x (P(x) \vee Q(x))$

13. i)

a) Solution: There is some student in your class who has sent a message to some student in your class.

b) Solution: There is some student in your class who has sent a message to every student in your class.

c) Solution: Every student in your class has sent a message to at least one student in your class.

d) Solution: There is a student in your class who has been sent a message by every student in your class.

e) Solution: Every student in your class has been sent a message from at least one student in your class.

f) Solution: Every student in the class has sent a message to every student in the class.

ii)

- | | |
|--|--|
| a) Solution: At least one student has taken class y. | b) Solution: At least one student has taken all classes. |
| c) Solution: Every student has taken at least one class. | d) Solution: At least one class has all students. |
| e) Solution: Every class has at least one student. | f) Solution: All students are taking all classes. |

14. i)

- a) Solution: Not a propositional Function since values are not provided.
b) Solution: Propositional Function. Truth Value: True
c) Solution: Not a propositional Function since values are not provided.
d) Solution: Not a propositional Function since values are not provided.
e) Solution: Propositional Function. Truth Value: False
f) Solution: Propositional Function. Truth Value: True

ii)

- | | | | |
|--------------------|-------------------|--------------------|-------------------|
| a) Solution: False | b) Solution: True | c) Solution: False | d) Solution: True |
| e) Solution: False | f) Solution: True | | |

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|-------------------------|--------------------|-------------------|-------------------|
| iii) a) Solution: False | b) Solution: False | c) Solution: True | d) Solution: True |
|-------------------------|--------------------|-------------------|-------------------|

- | | |
|-----------------------|-------------------|
| iv) a) Solution: True | b) Solution: True |
|-----------------------|-------------------|

15. i)

- | | | |
|----------------------------|-------------------------------------|---------------------------|
| a) Solution: Addition | b) Solution: Simplification | c) Solution: Modus ponens |
| d) Solution: Modus Tollens | e) Solution: Hypothetical Syllogism | |

ii)

a) Solution:

Converting to logical notation:

t = today is Tuesday

m = I have a test in Mathematics

e = I have a test in Economics

s = My Economics Professor is sick.

So the argument is:

$t \rightarrow (m \vee e)$

$s \rightarrow \neg e$

$t \wedge s$

$\therefore m$

- | | |
|-------------------------------|-------------------------------------|
| 1. $t \wedge s$ | premise |
| 2. t | from (1) by simplification Law |
| 3. $t \rightarrow (m \vee e)$ | premise |
| 4. $m \vee e$ | from (2) and (3) by modus ponens |
| 5. s | from (1) by simplification Law |
| 6. $s \rightarrow \neg e$ | premise |
| 7. $\neg e$ | from (5) and (6) by modus ponens |
| 8. $e \vee m$ | from (4) by commutative law |
| 9. m | from (7) and (8) by elimination Law |

b)

Solution:

$$P \rightarrow q$$

$$q \rightarrow r$$

$$p \rightarrow r$$

(Hypothetical syllogism)

$$P \rightarrow r$$

$$r \rightarrow \neg s$$

$$P \rightarrow \neg s$$

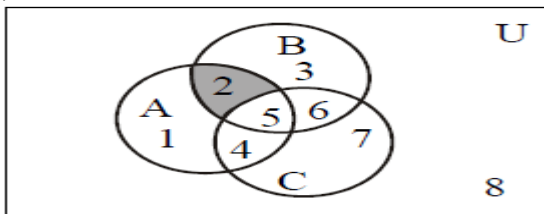
(Hypothetical syllogism)

Conclusion: $p \rightarrow \neg s$; if Ali is a lawyer then he does not like chocolates.

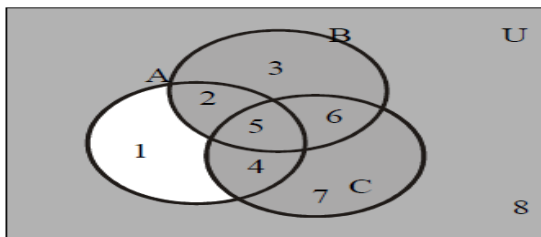
c) Solution: There is only single premise, hence no conclusion can be drawn.

16. i)

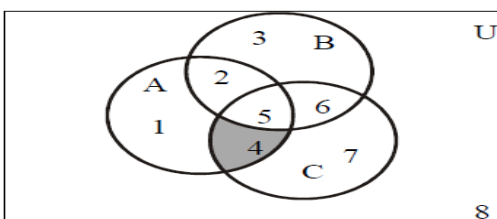
a) Solution:



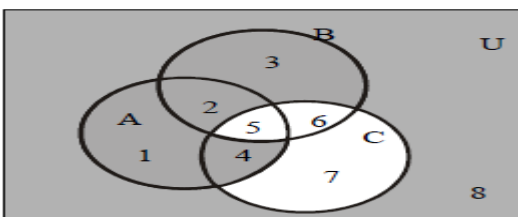
b) Solution:



c) Solution:



d) Solution:



- ii)
- a) Solution: A is a subset of B because all elements of A are also in B. A is not a proper subset of B because A and B have the same elements.
- b) Solution: Proper subsets of C include: $\{\}, \{2\}, \{3\}, \{5\}, \{7\}, \{2,3\}, \{2,5\}, \{2,7\}, \{3,5\}, \{3,7\}, \{5,7\}$.
- c) Solution: Subsets of L are: $\{\}, \{1\}, \{2\}, \{3\}, \{1,2\}, \{1,3\}, \{2,3\}, \{1,2,3\}$. Power set of L is: $\{\{\}, \{1\}, \{2\}, \{3\}, \{1,2\}, \{1,3\}, \{2,3\}, \{1,2,3\}\}$
- d) Solution: If set M has k elements, its power set will have 2^k elements.
- e) Solution: Set N = $\{11, 13, 17, 19\}$. Power set of N has $2^4 = 16$ subsets.
- f) Solution: $E \times F = \{(1,a), (1,b), (1,c), (1,d), (2,a), (2,b), (2,c), (2,d), (4,a), (4,b), (4,c), (4,d)\}$

17. i)

a) Solution:

$$\begin{aligned}
 &= (A \cap \overline{(A \cap B)}) \cap \overline{(B \cap \overline{(A \cap B)})} & (A - B) &= A \cap \bar{B} \\
 &= (A \cap B) \cap \overline{((A \cap B) \cap (A \cap B))} & \text{Associative Law} \\
 &= (A \cap B) \cap \overline{(A \cap B)} & (A \cap A) &= A \\
 &= \emptyset & (A \cap \bar{A}) &= \emptyset
 \end{aligned}$$

b) Solution:

$$\begin{aligned}
 \text{LHS} &= (A - B) \cup (A \cap B) \\
 &= (A \cap B^c) \cup (A \cap B) & \text{(Alternative representation for set difference)} \\
 &= A \cap (B^c \cup B) & \text{Distributive Law} \\
 &= A \cap U & \text{Complement Law} \\
 &= A & \text{Identity Law} \\
 &= \text{RHS} & \text{(proved)}
 \end{aligned}$$

c) Solution:

$$\begin{aligned}
 \text{LHS} &= A - (A - B) \\
 &= A - (A \cap B^c) & \text{Alternative representation for set difference} \\
 &= A \cap (A \cap B^c)^c & \text{Alternative representation for set difference} \\
 &= A \cap (A^c \cup (B^c)^c) & \text{DeMorgan's Law} \\
 &= A \cap (A^c \cup B) & \text{Double Complement Law} \\
 &= (A \cap A^c) \cup (A \cap B) & \text{Distributive Law} \\
 &= \emptyset \cup (A \cap B) & \text{Complement Law} \\
 &= A \cap B & \text{Identity Law} \\
 &= \text{RHS} & \text{(proved)}
 \end{aligned}$$

d) Solution:

$$\begin{aligned}
 (B^c \cup (B^c - A))^c &= (B^c \cup (B^c \cap A^c))^c \\
 &\text{Alternative representation for set difference} \\
 &= (B^c)^c \cap (B^c \cap A^c)^c & \text{DeMorgan's Law} \\
 &= B \cap ((B^c)^c \cup (A^c)^c) & \text{DeMorgan's Law} \\
 &= B \cap (B \cup A) & \text{Double Complement Law} \\
 &= B & \text{Absorption Law}
 \end{aligned}$$

ii)

a) Solution:

Total number of apples, $n(A) = 100$

Number of apples with bruises, $n(B) = 15$

Number of apples with worms or bruises,

$$n(W \cup B) = n(W) + n(B) - n(W \cap B) = 20 + 15 - 10 = 25$$

$$\text{Number of apples can be sold} = n(A) - n(W \cup B) = 100 - 25 = 75.$$

Number of apples with worm, $n(W) = 20$

Number of apples with both worms and bruises, $n(W \cap B) = 10$

b) Solution:

Total number of students, $n(\mu) = 1000$

Number of Software Engineering students, $n(SE) = 450$

Number of students who like either of them,

$$n(CS \cup SE) = n(CS) + n(SE) - n(CS \cap SE) = 450 + 350 - 100 = 700$$

$$\text{Number of students who like neither} = n(\mu) - n(CS \cup SE) = 1000 - 700 = 300$$

c) Solution:

Let the set of students who like mixed berry be M, those who like tiramisu be T and those who like Irish cream be I. Then, by the inclusion-exclusion principle, the number of students who like at least one of the flavors is

$$\begin{aligned} |M \cup T \cup I| &= |M| + |T| + |I| - |M \cap T| - |T \cap I| - |M \cap I| + |M \cap T \cap I| \\ &= 78 + 32 + 57 - 16 - 21 - 13 + 5 = 122 \end{aligned}$$

Now there are an additional 14 who like none of these three flavors, so the total number of students surveyed was $122 + 14 = 136$.

d) Solution: L.H.S

$$\begin{aligned} &= \{(x, y) : (x \in A) \wedge (y \in B \cap C)\} && \text{(def. of } \times) \\ &= \{(x, y) : (x \in A) \wedge (y \in B) \wedge (y \in C)\} && \text{(def. of } \cap) \\ &= \{(x, y) : (x \in A) \wedge (x \in A) \wedge (y \in B) \wedge (y \in C)\} && (P = P \wedge P) \\ &= \{(x, y) : ((x \in A) \wedge (y \in B)) \wedge ((x \in A) \wedge (y \in C))\} && \text{(rearrange)} \\ &= \{(x, y) : (x \in A) \wedge (y \in B)\} \cap \{(x, y) : (x \in A) \wedge (y \in C)\} && \text{(def. of } \cap) \\ &= (A \times B) \cap (A \times C) && \text{(def. of } \times) \end{aligned}$$

iii)

i) Solution: $\{1, 1, 1, 1, 0, 0, 0, 0\}$

ii) Solution: $\{0, 0, 0, 0, 0, 1, 1, 0\}$

iii) Solution: $\{1, 0, 0, 0, 1, 0, 1, 0\}$

b) i) Solution: $\{SE, AI, FT\}$

ii) Solution: $\{CS, SE, EE, BA, FT\}$

iii) Solution: $\{CS, AI, EE, FT\}$

c) Solution: $\{CS, SE, AI, CY, BA, FT\} = \{1, 1, 1, 1, 0, 1, 1, 0\}$

Solution: $\{CS, SE, AI, CY, EE, FT\} = \{1, 1, 1, 1, 1, 0, 1, 0\}$

Solution: $\{CS, SE, AI, EE, FT\} = \{1, 1, 1, 0, 1, 0, 1, 0\}$

Solution: $\{CS, EE, FT\} = \{1, 0, 0, 0, 1, 0, 1, 0\}$

Solution: $\{CS\} = \{1, 0, 0, 0, 0, 0, 0, 0\}$

Solution: $\{AI, FT\} = \{0, 0, 1, 0, 0, 0, 1, 0\}$

Solution: $\{CS, SE, AI, CY, FT\} = \{1, 1, 1, 1, 0, 0, 1, 0\}$

iv)

a) Solution: $A = \{(0, 8), (1, 4)\}$

$B = \{(a, 7), (b, 6)\}$

$C = \{(1, 2), (2, 4), (3, 3)\}$

$D = \{(ABC, 3), (BCA, 2), (ACB, 1)\}$

$E = \{(0, 4), (1, 6)\}$

$F = \{(a, 6), (b, 3), (c, 3)\}$

b)

Solution: $\{(0, 8), (1, 4), (a, 7), (b, 6)\}$

Solution: $\{(1, 2)\}$

Solution: $\{(a, 7), (b, 6), (c, 3)\}$

Solution: $\{(0, 4), (1, 6), (2, 4), (3, 3)\}$

Solution: $\{(a, 6), (b, 3)\}$

Solution: $\{(0, 8), (1, 6), (2, 4), (3, 3)\}$

Solution: $\{(ABC, 6), (BCA, 4), (ACB, 2)\}$

18.

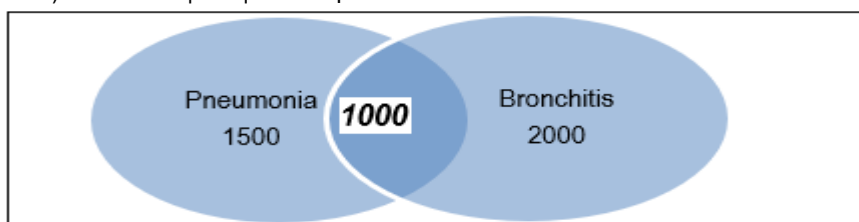
(i)

a) Solution: B is a subset of A. C is subset of A and D.

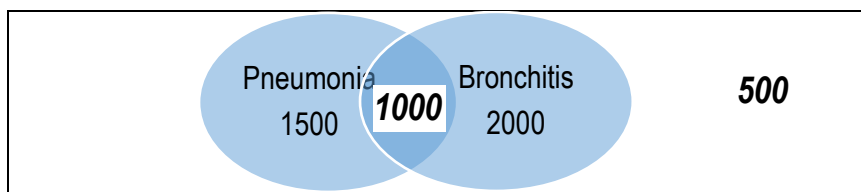
b) Solution: $n|A| = 3, n|B| = 2, n|C| = 2, n|D| = 3$.

(ii)

a) Solution: $n|A \cup B| = 4500$ patient



b) Solution: $U - n |A \cup B| = 500$ patients



(iii) Solution:

$$L.H.S = P - (Q \cap R)$$

$$= P \cap (Q \cap R)^c$$

$$= P \cap (Q^c \cup R^c)$$

$$= (P \cap Q^c) \cup (P \cap R^c)$$

$$= (P - Q) \cup (P - R) \neq R.H.S$$

Hence, it's a disproof.

$$A - B = A \cap B^c$$

De-Morgan Law

Distributive Law

$$A - B = A \cap B^c$$

(iv) a) Solution: Since f is bijective function, hence f^{-1} exists. $f^{-1} = \{(10, a), (7, b), (2, c)\}$

Since g is not bijective function, hence g^{-1} not exists.

b) Solution:

$f \circ g$ exists, hence $f \circ g = \{(1, 7), (2, 10), (3, 10), (4, 7)\}$.

$g \circ f$ doesn't exist because the range of f is not a subset of the domain of g .

19.

(i) Solution:	a)	Domain: $\{a, b, c, d\}$	Co-domain: $\{a, b, c, d\}$	Range: $\{a, b, c, d\}$
	b)	Domain: $\{a, b, c, d\}$	Co-domain: $\{a, b, c, d\}$	Range: $\{b, c, d\}$
	c)	Domain: $\{a, b, c, d\}$	Co-domain: $\{a, b, c, d\}$	Range: $\{b, c, d\}$
	d)	Domain: $\{a, b, c, d\}$	Co-domain: $\{a, b, c, d\}$	Range: $\{a, b, c, d\}$

(ii) Solution:

a) Injective and Bijective

b) Neither

c) Neither

d) Injective and Bijective

(iii) Solution:

$$a) f^{-1}(b) = a, f^{-1}(a) = b, f^{-1}(c) = c, f^{-1}(d) = d$$

b) Not exists

$$c) f^{-1}(c) = a, f^{-1}(a) = b, f^{-1}(b) = c, f^{-1}(d) = d$$

d) Not exists

20. a)

$$(i) \text{ Solution: } f(S) = \{1, 0, 0, 0, 1, 3\} \quad (ii) \text{ Solution: } f(S) = \{0, 0, 1, 3, 5, 8\}$$

$$(iii) \text{ Solution: } f(S) = \{0, 8, 16, 40\}$$

$$(iv) \text{ Solution: } f(S) = \{1, 12, 33, 65\}$$

b)

$$(i) \text{ Solution: } 1$$

$$(ii) \text{ Solution: } 0$$

$$(iii) \text{ Solution: } 0$$

$$(iv) \text{ Solution: } -1$$

$$(v) \text{ Solution: } 3$$

$$(vi) \text{ Solution: } -1$$

$$(vii) \text{ Solution: } 2$$

$$(viii) \text{ Solution: } 1$$

$$c) \text{ Solution: } \left\lfloor -\frac{3}{4} \right\rfloor = -\left\lceil \frac{3}{4} \right\rceil \rightarrow -1 = -1$$

$$\left\lceil -\frac{3}{4} \right\rceil = -\left\lfloor \frac{3}{4} \right\rfloor \rightarrow 0 = 0$$

Hence, It's a disproof.

$$21. a) \text{ Solution: } (f \circ g)(a) = f(g(a)) = f(3a + 2) = 2(3a + 2) + 3 = 6a + 7.$$

$$(g \circ f)(a) = g(f(a)) = g(2a + 3) = 3(2a + 3) + 2 = 6a + 11.$$

b) Solution: f and g are Injective.

c) Solution: Hence f and g are not invertible.

22. Solution:

a) $f(x) = x + 5$

b) $g(x) = 1.07x$

c) $f(g(x))$ means that the 7% will be applied to the original score and then 5 points will be added to that new score.

d) $f(g(75)) = f(80.25) = 85.25$

e) $g(f(x))$ means that the 5 points will be added to the original score and then the 7% increase will be applied to that new score.

f) $g(f(75)) = g(80) = 85.6$

g) No